Machine Intelligence 1

3.1 Uncertainty and Inference

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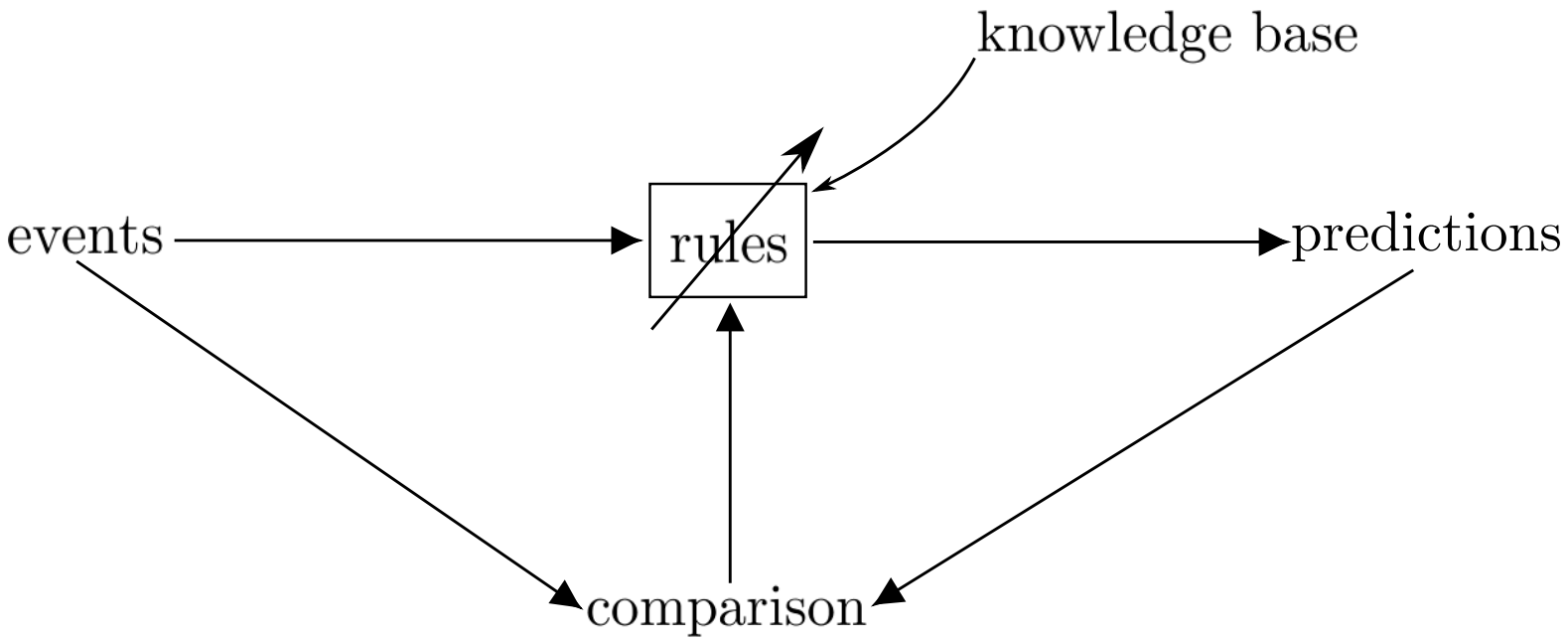
3.1.1 Degrees of Belief

Reasoning and learning



What about logic?

1. $\forall p : \text{symptom}(p, toothache) \Rightarrow \text{disease}(p, cavity)$
   - rule is wrong: not all patients with toothache have cavities
2. $\forall p : \text{symptom}(p, toothache) \Rightarrow$
   $\text{disease}(p, cavity) \vee \text{disease}(p, gum_disease) \vee \ldots$
   - almost unlimited list of possible causes
3. $\forall p : \text{disease}(p, cavity) \Rightarrow \text{symptom}(p, toothache)$
   - rule is wrong: not all cavities cause pain

- How can we make decisions when we are never 100% sure?

Degrees of belief (1)

- Proposition H

$$
\begin{aligned}
P(H) : H \rightarrow [0,1] \qquad & \text{assignment of numbers} \\
P(H) = 0 \qquad & H \text{ is false} \\
P(H) = 1 \qquad & H \text{ is true} \\
0 < P(H) < 1 \qquad & \text{quantifies our degree of belief}
\end{aligned}
$$

- $P(H)$ obeys the laws of probability theory
- but: no justification via repeated observations

E. T. Jaynes: *Probability theory – the logic of science* (2003)

Degrees of belief (2)

Application to betting agents de Finetti (1937)

"If Agent 1 expresses a set of degrees of belief that violate the axioms of probability theory, then there is a combination of fair bets by Agent 2 that guarantees that Agent 1 will loose money all the time."

- Agent 1 has inconsistent beliefs
  - which, e.g., violate $P(A \vee B) = P(A \wedge B) + P(\neg A \wedge B) + P(A \wedge \neg B)$
- Agent 2 can devise a fair bet such that Agent 1 always looses, no matter the outcome of A and B.

Agent 1		Agent 2		Outcome for Agent 1			
Proposition	Belief	Bet	Stakes	$A \wedge B$	$A \wedge \neg B$	$\neg A \wedge B$	$\neg A \wedge \neg B$
A	0.4	A	4 to 6	-6	-6	4	4
B	0.3	B	3 to 7	-7	3	-7	3
$A \vee B$	0.8	$\neg(A \vee B)$	2 to 8	2	2	2	-8
				-11	-1	-1	-1

(see blackboard)

3.1.2 The Description of the World

Random variables and their domains

- Boolean variables, e.g. $cavity \in \{\text{true}, \text{false}\}$
  - proposition: $cavity = \text{true}$
- discrete ordinal variables, e.g. $weather \in \{\text{sunny}, \text{rainy}, \text{cloudy}\}$
  - proposition: $weather = \text{sunny}$
- continuous variables, e.g. $temperature \in \mathbb{R}_0^+$
  - proposition: $temperature \in [290K, 291K]$
- description of the world: complete set of all variables

Atomic events

- atomic event: an assignment of values to all random variables
  - atomic events are mutually exclusive
  - set of atomic events must be exhaustive
- general proposition: disjunction of atomic events
  - e.g. $cavity =$ true is equivalent to:
    $(cavity =$ true $\wedge\, toothache =$ false $\wedge\, catch =$ false$) \vee$
    $(cavity =$ true $\wedge\, toothache =$ true $\wedge\, catch =$ true$) \vee \ldots$

Prior (unconditional) probabilities

- probabilities of atomic events
  - e.g. $P(cavity = \text{true} \wedge toothache = \text{true} \wedge catch = \text{true}) = 0.108$

- knowledge base:
  set of unconditional probabilities for all atomic events

	$toothache = \text{true}$		$toothache = \text{false}$	
	$catch = \text{true}$	$catch = \text{false}$	$catch = \text{true}$	$catch = \text{false}$
$cavity = \text{true}$	0.108	0.012	0.072	0.008
$cavity = \text{false}$	0.016	0.064	0.144	0.576

$$P(toothache, cavity, catch)$$

- probability of a proposition = sum of probabilities of all contributing atomic events ("marginalization")

3.1.3 Inference

Nomenclature

- Random variables (e.g. X, Y) are written in uppercase letters.
- Possible values are written in corresponding lowercase letters (e.g. x, y)
- Probability of an event (e.g., a variable assuming a value):

$$P(X = x) \mathrel{\widehat{=}} P(x), \text{ where } \sum_x P(x) = 1$$

Conditional probabilities

- probability of an event x given observations $\underline{\mathbf{e}} = (e_1, e_2, \ldots)$
  - X: query variable
  - E_j: evidence variables
  - Y_j: unobserved variables

$$P(x|\underline{\mathbf{e}}) = \frac{P(x, \underline{\mathbf{e}})}{P(\underline{\mathbf{e}})}$$

Computing conditional probabilities

- probability of an event x given observations $\underline{\mathbf{e}} = (e_1, e_2, \ldots)$
  - X: query variable
  - E_j: evidence variables
  - Y_j: unobserved variables

$$P(x|\underline{\mathbf{e}}) = \frac{P(x,\underline{\mathbf{e}})}{P(\underline{\mathbf{e}})} = \overbrace{\alpha\, P(x,\underline{\mathbf{e}})}^{\text{normalization}} = \overbrace{\alpha \sum_{\underline{\mathbf{y}}} P(x,\underline{\mathbf{e}},\underline{\mathbf{y}})}^{\text{marginalization}}$$

- normalization α can be computed...
  - ...explicitly by $\frac{1}{\alpha} = P(\underline{\mathbf{e}}) = \sum_{x,\underline{\mathbf{y}}} P(x,\underline{\mathbf{e}},\underline{\mathbf{y}})$
  - ...implicitly by ensuring $\sum_x P(x|\underline{\mathbf{e}}) = 1$

The effect of evidence

	$toothache =$ true		$toothache =$ false	
	$catch =$ true	$catch =$ false	$catch =$ true	$catch =$ false
$cavity =$ true	0.108	0.012	0.072	0.008
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- calculate $P(cavity)$ (see blackboard)

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$cavity =$ true	$cavity =$ false
0.2	0.8

- calculate $P(cavity \mid toothache = \text{true})$ (see blackboard)

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- calculate $P(cavity)$

$cavity =$ true	$cavity =$ false
0.2	0.8

- calculate $P(cavity \mid toothache = \text{true})$

$cavity =$ true	$cavity =$ false
0.6	0.4

Bayes' theorem

$$P(X|Y)\,P(Y) \;=\; P(Y,X) \;=\; P(Y|X)\,P(X) \qquad \text{(product rule)}$$

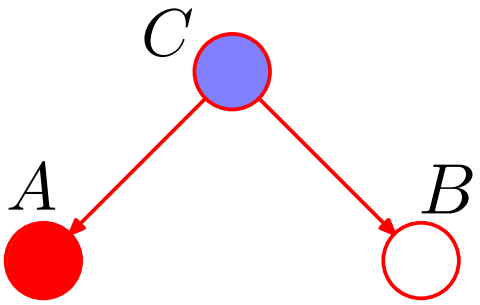
$$P(Y|X) \;=\; \frac{P(X|Y)\,P(Y)}{P(X)} \;=\; \alpha\,P(X|Y)\,P(Y) \qquad \text{(Bayes' theorem)}$$

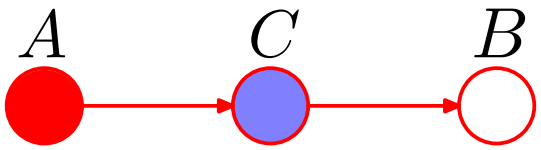
Computational complexity

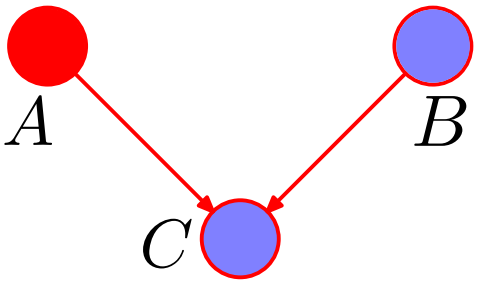
- Inference requires the complete table of prior (unconditional) probabilities.
- N binary random variables X_i:
  - table of joint probabilities has 2^N entries
  - summation over approx. 2^N entries for inference
  - $N = 100 \Rightarrow 2^N \approx 1.3 \cdot 10^{30}$
- This framework does not scale: additional assumptions are needed.

3.1.4 Conditional Independence

Dependencies







Conditional independence

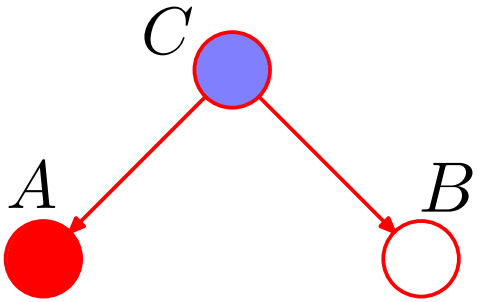
Two random variables X and Y are conditionally independent given Z if:

$$P(X, Y|Z) \quad = \quad P(X|Z) \;\; P(Y|Z)\,.$$

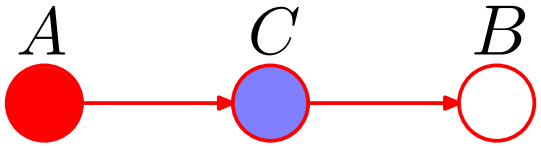
We write $X \perp Y \mid Z$.

Statistical vs. conditional independence

$$X \perp Y \mid Z \quad \Leftrightarrow \quad P(X,Y|Z) = P(X|Z)\ P(Y|Z)$$



$A \perp B \mid C$
$A \not\perp B \mid \emptyset$



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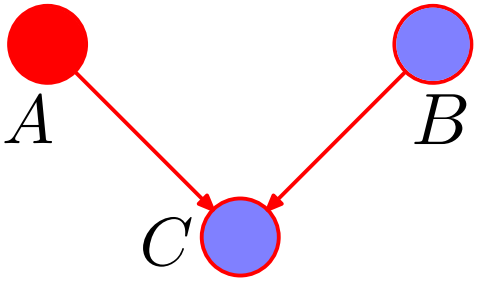

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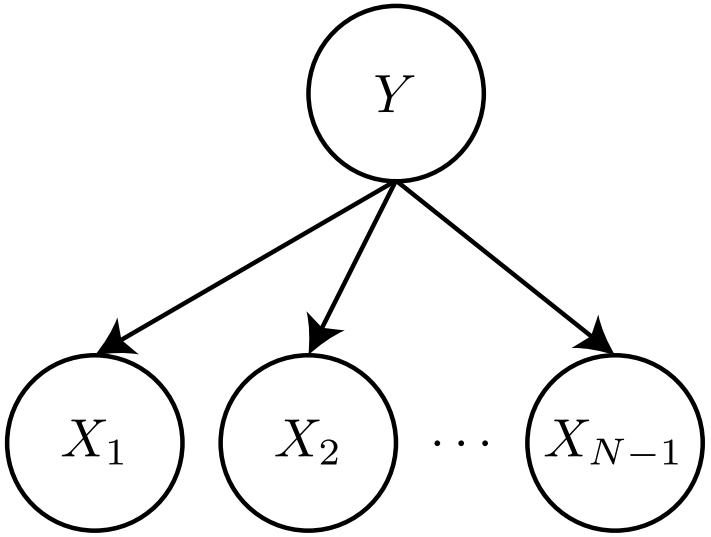
(see blackboard)

The Naïve Bayes ansatz



- set of N binary variables:
  X_i with $i = 1, \ldots, N-1$ and Y

- assumption $X_i \perp X_j \,|\, Y, \; \forall i, j$

$$\underbrace{P(X_1, X_2, \ldots, X_{N-1}, Y)}_{2^N-1 \text{ independent table entries}} \quad = \quad \underbrace{P(Y) \prod_{i=1}^{N-1} P(X_i|Y)}_{1+2(N-1)=2N-1 \text{ independent table entries}}$$

- reduction of computational complexity from $O(2^N)$ to $O(N)$

References I

Bruno de Finetti. La prévision: ses lois logiques, ses sources subjectives. In *Annales de l'institut Henri Poincaré*, volume 7, pages 1–68, 1937. Translation reprinted in H.E. Kyburg and H.E. Smokler (eds.) (1980), Studies in Subjective Probability, 2nd edn (pp. 53-118). New York: Robert Krieger.